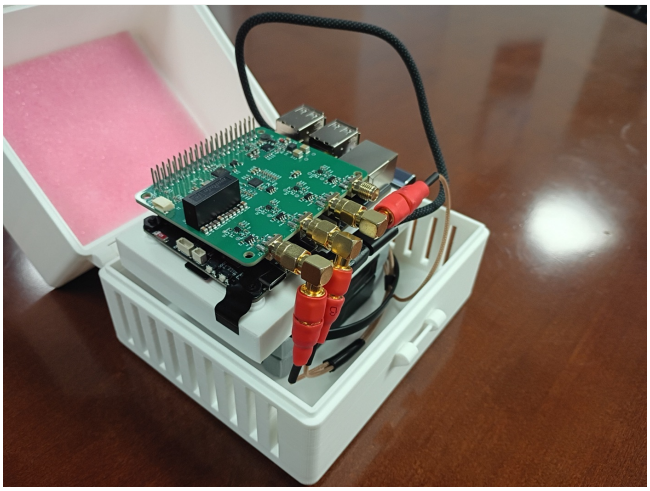
gLOWCOST detector quick guide

ESP32-P4 field start-up and Wi-Fi setup

Start a run

1. Insert a FAT32-formatted microSD card before power-up.
2. Connect the detector to the USB power bank. It starts automatically.
3. Leave the detector still while the FPGA, DAC, and high voltage are configured.
4. Wait through the 10 second HV settling period. Counting and SD logging then begin.



Detector opened for setup

Open the detector page

Turn mobile data off so the phone stays connected to the detector network.

Wi-Fi	Password	Web page
MuonReadout	**glowcost**	**192.168.4.1**

Before the measurement

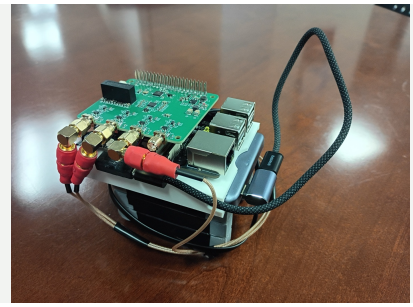
- [ ] SD Card shows Ready.
- [ ] Browser time has synced to the detector.
- [ ] One-minute coincidence counts look reasonable.
- [ ] Flight number, location, or run label is set if needed.
- [ ] Wi-Fi is off before the quiet measurement period.

Five-minute Wi-Fi window

A countdown is shown on the web page. Wi-Fi turns off five minutes after startup, even if a phone remains connected. Select **Keep Wi-Fi On** only when a longer setup session is needed. The choice lasts until the next power cycle.



Closed case



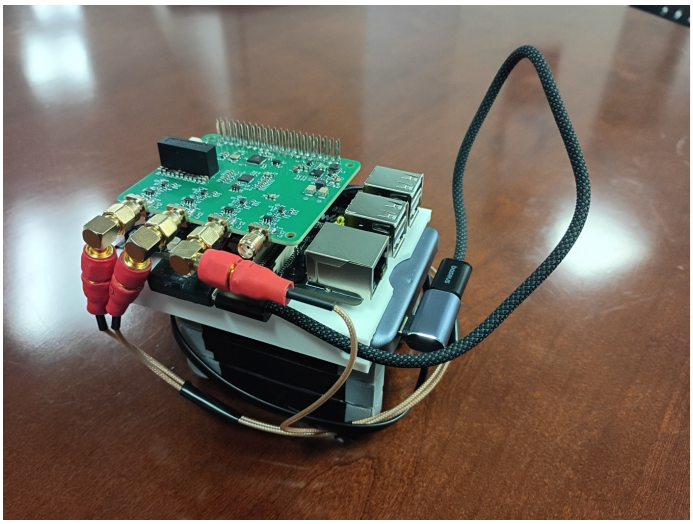
Powered detector



5000 mAh power bank

Wired USB-UART monitor

Use the Mac dashboard while detector Wi-Fi remains off



Keep the detector powered from the power bank during monitoring.

Connect the Mac

1. Leave the normal USB-C power input connected to the power bank.
2. Connect the separate USB-UART Type-C port on the ESP32-P4 board to the Mac.
3. Start the Mac monitor and keep its Terminal window open.
4. Open **http://127.0.0.1:8080** if the browser does not open automatically.

Start the dashboard

Packaged Mac monitor

Unzip **Muon-UART-Monitor-macOS.zip**, then double-click **Start Muon UART Monitor.command**. If macOS blocks it, right-click the file and choose **Open**.

From a repository checkout

```
python3 tools/uart_web_monitor.py
--open-browser
```

The wired page provides

- [ ] Latest 10 count lines, with the newest line highlighted.
- [ ] Automatic time sync from the Mac browser.
- [ ] SD, FPGA, HV, counting, and environment status.
- [ ] Run-label changes and SD file downloads.
- [ ] Protected Wi-Fi shutdown and FPGA reflash controls.

The detector keeps running

Keep the power bank connected as the detector's main supply. Closing the Mac program or unplugging the UART cable does not reset the detector or stop its counters and SD logging.

If the page does not open

Confirm Python 3 is installed and a serial device appears with **ls /dev/cu.usbmodem***. Close **screen**, miniterm, or any other program using the UART port, then restart the monitor.

Time and filenames

Opening the page sends the Mac's local time to the detector. On the first valid sync after startup, the active muon and environment files are renamed with that timestamp.

Wi-Fi is not required for the wired dashboard.